



CSIS3744

Computer Networks

(NQF level 7 - 16 credits)

Dept. of Computer Science &
Informatics

Compiled by
Dr O. Oriola

2026

Faculty of **Natural and
Agricultural Sciences**

Fakulteit **Natuur- en
Landbouwetenskappe**

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1. WELCOMING

Dear student,

Welcome to CSIS3744, Computer Networks! I am Dr Oluwafemi Oriola and will be responsible for teaching this module. Here is all the information that you will need to contact me, as well as other relevant staff in the Department of Computer Science and Informatics.

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Head of Department:	Prof. Eduan Kotzé
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2. MODULE OBJECTIVE

In this section, I will provide you with a short overview of what will be expected from you to be successful in your learning in this module.

The aim of CSIS3744 is to provide you with an overview of network concepts. Aspects that are covered are network architecture, network technologies, coupling techniques, internetwork concepts, end-to-end protocols, security, standards and models, transmission basics, and network applications.



3. MODULE OUTCOMES

After successful completion of this module, you should be able to:

- Distinguish between the fundamental network types,
- Conceptualise and explain network communications by means of the OSI model, TCP/IP model and TCP/IP protocols,
- Describe and explain wired as well as wireless LAN and WAN topologies, transmission methods, network media, access methods and Ethernet standards,
- Identify and distinguish between network hardware and explain switching and routing methods,
- Explain IP addressing, implement subnetting and troubleshoot network problems,
- Explain and discuss network management aspects and network security techniques,
- Set up networks with virtual machines and simulation tools.



4. LEARNING UNITS

In addressing the module outcomes above, the following learning units will be covered in this module.

Learning unit 1:	Introduction to Networking
Learning unit 2:	Infrastructure and Documentation
Learning unit 3:	Addressing
Learning unit 4:	Protocols
Learning unit 5:	Cabling
Learning unit 6:	Wireless Networking
Learning unit 7:	Network Architecture
Learning unit 8:	Network Segmentation
Learning unit 9:	Wide Area Networking
Learning unit 10:	Risk Management
Learning unit 11:	Access Control
Learning unit 12:	Performance and Recovery



5. PREREQUISITES

Students must have successfully completed CSIS1624 or CSIE1606 before they can register for CSIS3744. For more information, please consult the faculty rule book, available for download on the departmental website: www.ufs.ac.za/csi.



6. CREDITS AND NOTIONAL LEARNING HOURS

'Notional learning hours' indicate the amount of learning time taken by the 'average' student to achieve the specified learning outcomes of the course unit or programme. This includes all learning relevant to achieving the learning outcomes, e.g., directed study, essential practical work, group work, private study, preparation and assessment.

This module has 16 credits, requiring 160 notional learning hours from students.



7. CONTACT SESSIONS

This module will make use of blended learning. Students are expected to attend two theoretical sessions (lectures) and one practical session per week. The extra one lecture session per week will be utilised if needed. To pass this module a minimum of 80% class attendance is required. Students can receive an *incomplete* for the module if the minimum class attendance is not met.

SESSION	DAY & TIME	VENUE
Lecture 1	Monday: 09h00 – 10h00	W 202
Lecture 2	Tuesday: 17h00 – 18h00	Stabilis 4
Lecture 3	Thursday: 17h00 – 18h00	Stabilis 4
Practical (Group A)	Monday: 14h10 – 17h00	WWG 223
Practical (Group B)	Monday: 17h10 – 20h00	WWG 316

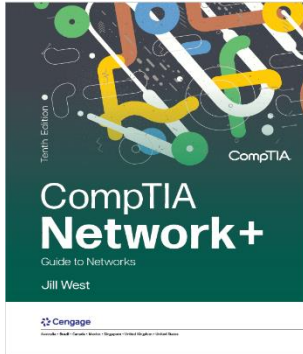
The scheduled practical session is there for assessment and assignment purposes and therefore, not compulsory. You can do your practical assignments on your own device or make use of the computer labs if you do not have access to your own device.

It is the responsibility of each student to ensure that they receive all the information provided in connection with this module during the theoretical and practical sessions. This information will also be made available on Blackboard. Please note that only registered students will be allowed to attend the CSIS3744 contact sessions.



8. STUDY MATERIAL

The prescribed textbook in this module is:



CompTIA Network+ Guide to Networks, 10th Edition

Jill West

Cengage

2025

ISBN: 979-8-214-01279-7

In addition to the textbook, other study material and resources will be used as part of the teaching and learning in this module. This material, resources and activities will be made available on Blackboard throughout the semester. It is each student's responsibility to ensure that they obtain and prepare all study material in time.



9. ASSESSMENT

9.1. ASSESSMENT BREAKDOWN

Your semester mark (predicate) will be calculated as follows:

ASSESSMENT	TYPE	CONTRIBUTION: SEMESTER MARK	DATE
Semester Test 1	Theoretical	35%	20 Aug 2026
Semester Test 2	Theoretical	35%	8 Oct 2026
Aegrotat Test	Theoretical	35%	19 Oct 2026
Assignments	Practical	20%	
Quizzes	Theoretical/Practical	10%	
Final Examination	Theoretical		Oct / Nov

9.2. SEMESTER TESTS

You must write both semester tests; otherwise, the module will be indicated as incomplete. After the result of the relevant semester test has been published, you will have a week (7 days) to query your results. Thereafter no changes will be made to your marks.

9.3. ASSIGNMENTS

All assignments are compulsory. No late submissions will be accepted (unless stated otherwise). Your module will be marked incomplete if you do not complete all the assignments to the satisfaction of the lecturer.

You may discuss the assignments with each other. However, you may only submit your own work. The ethical use of AI technologies is permitted without committing plagiarism. Students found guilty of any transgression in this regard will be severely penalised. Action will also be taken against students who help or allow other students to copy their work. The only exception to this rule is when assignments are classified as group assignments. (Also refer to the Departmental Disciplinary Code on Blackboard.)

It is your own responsibility to keep backup copies of your assignments. Lost or corrupted computer files, network problems, etc., cannot be used as an excuse for late submissions.

More detailed assignment information will be made available on Blackboard throughout the semester. It is each student's responsibility to ensure that he/she takes note of all the information and aligns him/herself with the expectations.

9.4. AEGROTAT TEST

The Aegrotat Assessment Policy of the Department of Computer Science and Informatics will be applied strictly. This policy is available on the Blackboard module website.

Please note:

- Should you qualify for the aegrotat assessment but fail to show up at the specified date, time and place, no additional module assessments will be drawn up.
- Participation in cultural activities, weddings, or transport problems, etc. are not valid reasons for absence during module assessments and will not justify qualification for an aegrotat assessment.
- No member of the Student Representative Council or House Committee may give you permission not to write a test or examination.

There will be only one aegrotat assessment, of which the date will be communicated later if applicable. This assessment will cover all the work of the semester.

9.5. INCOMPLETE

Failure to write/do any of the module assessments while not handing in a valid original or certified doctor certificate (within 48 hours after the assessment), will result in CSIS3744 being indicated as *incomplete* next to your student number. This implies that you will not be admitted to the final examination and that you will not be allowed to register for the module again (which could in effect mean the end of your studies at the UFS).

The module will be indicated as *incomplete* next to your student number in any of the following cases:

- If you fail to submit any compulsory assignment.
- If you fail to write any of the semester tests (or the aegrotat test instead of one of these tests).

9.6. PROMOTION

CSIS3744 is not a promotion module.

9.7. FINAL EXAMINATION

Only students with a semester mark of 45% or higher will be allowed access to the exam. The main and additional examinations will be in the format of a 3-hour theoretical written exam.

9.8. CALCULATION OF THE FINAL MARK

In terms of the rule book of the Faculty of Natural and Agricultural Sciences, the final mark of this module is calculated as follows: 50% of the semester mark plus 50% of the examination mark.

9.9. PASSING THE MODULE

A student should obtain a final mark of 50% or more in order to pass this module. In addition, a sub-minimum examination mark of 40% is required.



10. MODULE CALENDAR

The module stretches over a 14-week period, during which the following programme will be followed:

WEEK	MONTH	DATE	DESCRIPTION
1	Jul	13 – 17	Registration Orientation & Introduction to Networking
2	Jul	20 - 24	Introduction to Networking
3	July	27 – 31	Infrastructure and Documentation
4	Aug	3 – 7	Addressing
5	Aug	10 – 14	Protocols Submit Assignment 1: 10 Aug. 2026
6	Aug	17 – 21	Cabling Semester Test 1: 20 Aug. 2026
7	Aug	24 - 28	Wireless Networking
8	Aug/Sep	31 - 4	Network Architecture
9	Sep	7 - 11	Network Segmentation Submit Assignment 2: 7 Sept. 2026
10	Sep	14 - 18	Wide Area Networking
11	Sep	21 - 25	Risk Management Submit Assignment 3: 21 Sept. 2026
Recess (Academic Break and Holidays)			
12	Oct	6 - 9	Access Control Semester Test 2: 8 Oct. 2026
13	Oct	12 - 16	Performance and Recovery
14	Oct	19 – 23	Aegrotat Test: 19 Oct. 2026 Predicate Day: 23 Oct. 2026



11.UPDATING OF MODULE INFORMATION AND RESOURCES

The module schedule and syllabus (study material), either orally or in writing, can be updated or modified at any time by the lecturer or the Department of Computer Science and Informatics. It is the responsibility of students enrolled in this module to stay up to date with the schedule and syllabus. Announcements of this type can be given in face-to-face class sessions or communicated on Blackboard. It is recommended that students check Blackboard daily to keep up with the latest developments in CSIS3744. Any work discussed during contact sessions can be used for assessment purposes.